

Hassan Rehan

SENIOR SOFTWARE ENGINEER | CTO | NEXT.JS, NESTJS & AWS

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Portfolio: <https://hrmemon.github.io/hr-mern-portfolio/> | Startup: <https://parhako.com/>

Professional Summary

Senior Software Engineer and CTO with 5+ years of experience building high-performance web applications. Deeply specialized in the **JavaScript/TypeScript ecosystem (Next.js, NestJS)** and **AWS Cloud Architecture**. Transitioned from NoSQL to **PostgreSQL** for robust data modeling. Proven track record of delivering MVPs in record time, optimizing AI data pipelines, and debugging complex backend concurrency issues.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, SQL, Python.
- **Frontend:** Next.js, React.js, Tailwind CSS, Redux, Angular.
- **Backend:** Node.js, NestJS, Express.js, Prisma ORM, Django.
- **Databases:** PostgreSQL (Advanced), MongoDB, Redis, Bitnine.
- **AWS Cloud:** EC2, RDS, Lambda, S3, SQS, SNS, SES, CloudWatch.
- **DevOps & Tools:** Docker, CI/CD, Git, Jest.

Experience

Parhako | Remote

Co-Founder & CTO | Jan 2025 – Present

Founded and architected an AI-powered platform. Led the project from ideation through development to a successful go-to-market launch.

- **Full Stack Architecture:** Built a type-safe application using **Next.js** for the frontend and **Node.js/Express** for the backend.
- **Database & ORM:** Designed a complex relational schema using **PostgreSQL** and **Prisma**, prioritizing data integrity over NoSQL flexibility.
- **AI Integration & Optimization:** Integrated Generative AI APIs for content generation. Implemented aggressive **Regex-based data cleaning** and token optimization strategies, reducing API costs by 60%.
- **AWS Deployment:** Deployed a cost-efficient infrastructure using **EC2** for hosting and **S3** for asset storage. Configured **SES** for transactional emails.
- **Data Pipeline:** Built automated scripts to scrape, validate, and clean large datasets before feeding them into the production database.

TopDevSpace LLC | Remote

Senior Software Engineer | Aug 2023 – Dec 2024 & Oct 2025 – Dec 2025

Contract II: Rapid MVP Consultant (Oct 2025 – Dec 2025)

*Recalled by the client to architect and ship a critical product using **Next.js** and **NestJS**.*

- **Rapid MVP Delivery:** Successfully delivered a production-ready Full Stack MVP in just **4 sprints**.
- **Integrations:** Implemented **Stripe** for payments, **QuickBooks** for accounting synchronization, and integrated **LLM APIs** to power an intelligent chat interface.

Contract I: Core Senior Developer (Aug 2023 – Dec 2024)

Worked on complex architectural challenges across three distinct projects:

- **Django & HPC Optimization:** Managed data processing on **High-Performance Computing (HPC)** clusters. Identified a critical logical bug where auxiliary pipelines were triggering prematurely. Fixed the orchestration logic, resulting in a **70% reduction** in total processing time.
- **Microservices & Cloud:** Spearheaded a greenfield **Microservices** initiative. Successfully delivered the project while implementing **AWS** best practices and distributed system architecture.
- **Angular, Node & DevOps:** Diagnosed a severe bottleneck in the dev/QA cycle caused by inefficient **Monorepo deployment** times. Highlighted the infrastructure flaw to leadership and proposed a streamlined strategy to accelerate testing loops.

Incio | Remote

Full Stack Developer | Jan 2023 – April 2023

- Developed a business management SaaS platform using the **MERN Stack**.
- Integrated **Zoom and Google Meet APIs** to enable users to schedule and host meetings directly within the app.
- Built a custom invoicing module and customer management system to digitize client workflows.

Bitnine Global Inc. | Remote

Software Engineer

- Developed core web services using Bitnine's AgensGraph (RDB+GDB multi-model database).
- Worked on data visualization tools for graph technologies and Big Data integration.

Erays Technology | Hybrid

Software Engineering Intern | Apr 2022 – June 2022

Successfully managed a full-time internship while completing 3rd-year university coursework.

- Contributed to the flagship foot-mapping product using sensor data.
- Developed a heat-map visualization using data from 48 distinct pressure sensors, aiding in the technical analysis of foot posture.

Education

Bachelor of Engineering in Software Engineering

National University of Sciences and Technology (NUST)

Certificates

- **Advanced Node.js** – Scalable Architecture & System Design
- **Deep Learning Specialization** – DeepLearning.AI (TensorFlow, Neural Networks)
- **Generative Adversarial Networks (GANs)** – DeepLearning.AI (PyTorch)
- **Modern React with Redux** – Advanced State Management